

Exhibit CC-139

SAMSUNG

Product brief

Efficiently powering mobile innovation

S2FPC01 Power IC

Highlights

Efficient architecture to reduce power consumption

Fast transients for output voltage accuracy

Current capability with the best performance for fast data rate and high density DIMMs

DDR5 PMIC for client PC

The S2FPC01 is built on a 90nm process node optimized for small battery-powered devices, and comes in a smaller package. It has a more agile performance, and is tailored for use in desktop or laptop PCs.

Low power consumption

The S2FPC01 offers up to 91% power-efficiency. The asynchronous-based dual phase buck control scheme allows stepping down DC voltages from high to low with a fast transient response to changes in output load current. It efficiently regulates output voltage at near-constant levels. The control scheme also features both pulse width modulation and pulse frequency modulation methods which prevent delays and malfunctions during mode switching.

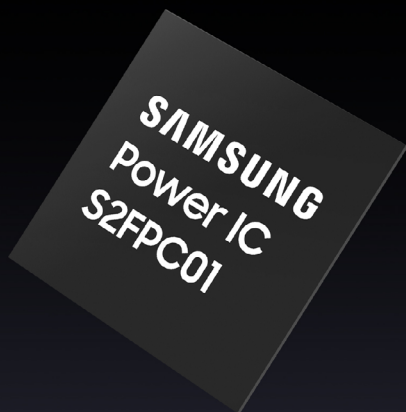
Optimized current capability

It has 3 bucks (SWA/SWB/SWC) and an LDO. The current capability of SWA and SWB is 4A and current capability of SWC is 1A. Dual phase mode (SWA+SWB) supports 8A. It is optimized for fast data rate and high density DIMMs for cost effectiveness.

JEDEC-compliant PMIC

The S2FPC01 fully supports JEDEC standard (version 1.0a).

(PMIC5100 Power Management IC Specification Rev1.0, JESD301-2)



Specifications

	S2FPC01 Power IC
VIN_BULK	4.25V ~ 5.5V
Power Management	Input UVLO/OVLO, Thermal Shut Down function Read/Write I2C/I3C for Register Control Interface up to 1MHz for I2C and 12.5MHz for I3C Error Log Count & Storage Built-in Nonvolatile Memory registers Dynamic Voltage Scaling controlled by I2C/I3C Programmable Soft Start & Soft Stop Time Current Limit, Over/Under voltage protection Power Meter
Current Capability	4A/4A/1A per each phase
SWA	Programmable output with 5mV step (0.6V ~ 1.435V) Switching Frequency : 0.75MHz
SWB	Programmable output with 5mV step (0.6V ~ 1.435V) Switching Frequency : 0.75MHz
SWC	Programmable output with 5mV step (1.5V ~ 2.135V) Switching Frequency : 0.75MHz
LD01	Programmable output voltage 1.7/1.8/1.9/2.0V Output current capacity of 25mA
LD02	Programmable output voltage 0.9/1.0/1.1/1.2V Output current capacity of 20mA

* All product specifications reflect internal test results and are subject to variations by user's system configuration. Actual performance may vary depending on use conditions and environment.
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